

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Fourth Semester B. Tech (CBCS) Degree Examinations**

Summer Semester July 2024  
**IS2001-1 – INTERNET AND WEB PROGRAMMING**  
 (IS)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

- In HTML, what is the purpose of the <meta> tag?**
  - To define metadata about an HTML document
  - To create hyperlinks
  - To insert media files
  - To define formatting styles
- What is the purpose of the HTML <nav> tag?**
  - It defines navigation links within a document
  - It creates a navigation bar
  - It specifies the background image of a page
  - It defines a new paragraph
- How do you link an external CSS file to an HTML document?**
  - <style> tag
  - <link> tag
  - <css> tag
  - <script> tag
- What is the purpose of the CSS property margin?**
  - Sets the spacing between lines of text
  - Sets the space outside the border of an element
  - Sets the space inside the border of an element
  - Sets the font size of an element
- What is the default value of the position property in CSS?**
  - absolute
  - relative
  - fixed
  - static
- What does the === operator in JavaScript check for?**
  - Equality in value
  - Equality in value and type
  - Assignment
  - Inequality in value
- How can you access an element by its ID in the DOM?**
  - document.getElementById('id')
  - document.getElementByName('id')
  - document.getElementById('id')
  - document.querySelector('#id')
- Which property is used to change the text content of an element?**
  - element.textContent
  - element.innerText
  - element.innerHTML
  - Both A and B
- What does the isset() function in PHP check for?**
  - Checks if a variable is set and is not NULL
  - Checks if a file exists
  - Checks if a function is defined
  - Checks if a session is active
- To fetch a row of data from a MySQL query result set in PHP, which function would you use?**
  - mysqli\_fetch\_row()
  - mysql\_fetch\_array()
  - fetch\_row()
  - pdo\_fetch()
- What is React.js?**
  - A back-end JavaScript framework
  - A front-end JavaScript library for building user interfaces
  - A database management system
  - A server-side scripting language
- Which package manager is commonly used with Node.js?**
  - Maven
  - npm
  - pip
  - Gradle

13. How can you include external modules in Node.js?  
 A) require() B) import  
 C) include() D) use()
14. Which status code represents a successful HTTP response in Node.js?  
 A) 200 B) 404  
 C) 500 D) 302
15. In MongoDB, what is a document?  
 A) A table B) A record  
 C) A JSON-like data structure D) A field
16. What does the find() method do in MongoDB?  
 A) Creates a new document B) Queries a collection for documents  
 C) Updates a document D) Deletes a document
17. In the context of CSS specificity, which selector has the highest specificity?  
 A) ID selector B) Class selector  
 C) Element selector D) Universal selector
18. Which value of the target attribute opens the linked document in a new browser window or tab?  
 A) \_self B) \_blank  
 C) \_parent D) \_top
19. Which of the following is a primary classification for NoSQL architectures?  
 A) Document databases B) Graph databases  
 C) Key-value databases D) All of the above
20. Which property is used to change the text color of an element?  
 A) color B) font-color  
 C) text-color D) text

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

#### Unit – I

1. a) Explain different states of anchor tag. Demonstrate list style type with user defined image logos.  
 b) Write a code in HTML that displays the HTML form as shown in below figure.

| Marks | BT* | CO* | PO* |
|-------|-----|-----|-----|
| 8     | L*2 | 1   | 1   |

CustomerID: \_\_\_\_\_

ProductID: \_\_\_\_\_

Gender  
☐ Male ☐ Female

Type of Product Beauty ▼

₹ \_\_\_\_\_  
 Electronics  
 Home Appliances  
 Clothing and Accessories

Complaint \_\_\_\_\_

Submit Reset

|   |    |   |   |
|---|----|---|---|
| 8 | L3 | 1 | 1 |
|---|----|---|---|

2. a) Use External style sheet to add the following CSS style specification
- Display the overflowed content when hovering over the element
  - Add link to the bottom and top of the page
  - Redirecting current page to another page after 5 seconds using HTML5 meta tag
  - Paragraph has to be enclosed in a border with groove and text font style italics, font color red.
- b) List three attributes of <form> element. Explain each of them with programming examples

8 L3 1 1  
8 L2 1 1

3. a) Explain <audio> and <video> tags. List at least 5 attributes with its usage.
- b) Write the HTML code to display the following table.

8 L2 1 1

*A test table with merged cells*

|         | Average |        | Red eyes |
|---------|---------|--------|----------|
|         | height  | weight |          |
| Males   | 1.9     | 0.003  | 40%      |
| Females | 1.7     | 0.002  | 43%      |

Also add the below CSS Properties.

- Style Male and Female details with different background-colors
- Change the background color when user places pointer over the rows

8 L3 1 1

#### Unit – II

4. a) Develop a HTML Form, which accepts any Mathematical expression. Write JavaScript code to Evaluates the expression and Displays the result.
- b) List any 4 super globals in PHP? Explain them by giving examples

8 L3 2 1  
8 L2 2 1

5. a) Write a PHP program to connect to the database. Create a table named "Emplyoe" with following fields (eno, ename, Designation, Salary). Insert 3 records of your choice. Display all the records inserted.
- b) What are the advantages of using React? Explain React state and props.

8 L3 3 1  
8 L2 3 1

6. a) Write an HTML page that contains a selection box with a list of 3 countries. When the user selects a country, its capital should be printed next to the list. Add CSS to customize the properties of the font of the capital (color, bold and font size).
- b) Write a React JS code to demonstrate how to locally manage a component's state.

8 L3 3 1  
8 L3 3 1

#### Unit – III

7. a) What is NodeJS? What are the advantages of NodeJS? Write a Node.js program utilizing the File System module to rename a file.

8 L3 4 1

- b) Create a MongoDB collection named 'games' and provide queries for the following operations:
- i) Add 5 games with Name, ID, and Rating properties
  - ii) Write a query that returns the 3 highest rated games.
  - iii) Sort the Games based on the names
- 8      L3      5      1
8. a) Write a Node.js script to establish an HTTP server that listens on specified ports and sends a response to clients upon receiving requests.
- 8      L3      4      1
- b) Give the structure of a MongoDB document. Explain the following functions with examples
- i) findOne()
  - ii) DeleteMany()
- 8      L2      5      1

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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